reproduces. All its interactions with other organisms and the nonliving environment are also part of what makes up its niche. A unique niche is an advantage for any animal or plant because this reduces competition with other species. For ecologists, a full knowledge of an organism's niche is vital to inform interventions to compensate for the environmental changes caused by habitat destruction and climate change.

The pioneer of the niche concept was Joseph Grinnell, a US biologist who studied a bird called the California Thrasher. In 1917, he published his observations, which showed how the bird fed and bred in the underbrush of a scrubby habitat known as chaparral, and how it escaped predators by running through the underbrush. The thrasher's camouflage, short wings, and strong legs were perfectly adapted for life in this environment. Grinnell saw the chaparral habitat as the thrasher's "niche." His idea also allowed for "ecological equivalence" in plants and animals, whereby species distantly related and living far apart could show similar adaptations, such as feeding habits, in similar niches. In the Australian outback, for instance, babbler bird species forage in the scrubby vegetation in a similar way to the unrelated thrasher. Grinnell also identified "vacant" niches—habitats that a species could potentially occupy, but where it was not present.